AI-Powered Workflow for Constructing Organic Materials Databases

Integrating Large Language Models for Literature Mining

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Presented by

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The Challenge: Data Scarcity in Materials Science

The Bottleneck

ML and data-driven methodologies have tremendous potential to revolutionize materials research, but progress is severely constrained by a critical shortage of available datasets.

Why Data is Scarce

- Experimental data is scattered across unstructured publications
- Information hidden in text, tables, and figures
- No centralized platforms for many research domains
- Manual extraction is labor-intensive and requires domain expertise

The Traditional Approach: Manual Data Extraction

Problems

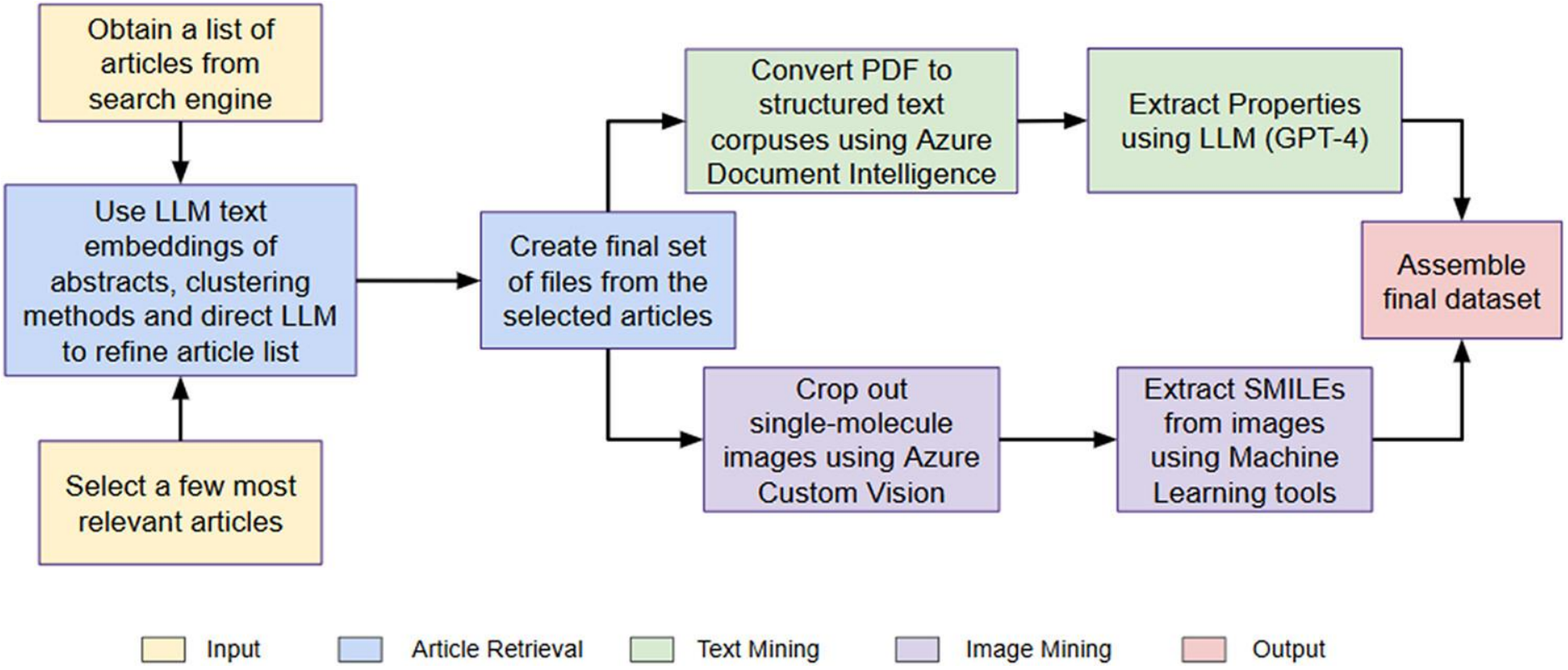
- Extremely labor-intensive and time-consuming
- Requires well-trained domain experts
- Not scalable across research domains
- Quickly becomes outdated

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YEAR

for one researcher to build a dataset of 1,203 materials from 503 papers

Solution: AI-Powered Automated Workflow



Three-Step Workflow

1

Article Retrieval

LLM embeddings, clustering, and direct queries to identify relevant papers

2

Text Mining

GPT-4 extracts materials properties from text and tables

3

Image Mining

Computer vision extracts molecular structures as SMILES notation

Step 1: Article Retrieval with LLM Embeddings

Four-Step Process

1. Keyword Search

Broad search via Semantic Scholar API

2. Text Embedding

Convert abstracts to high-dimensional vectors using OpenAI embeddings

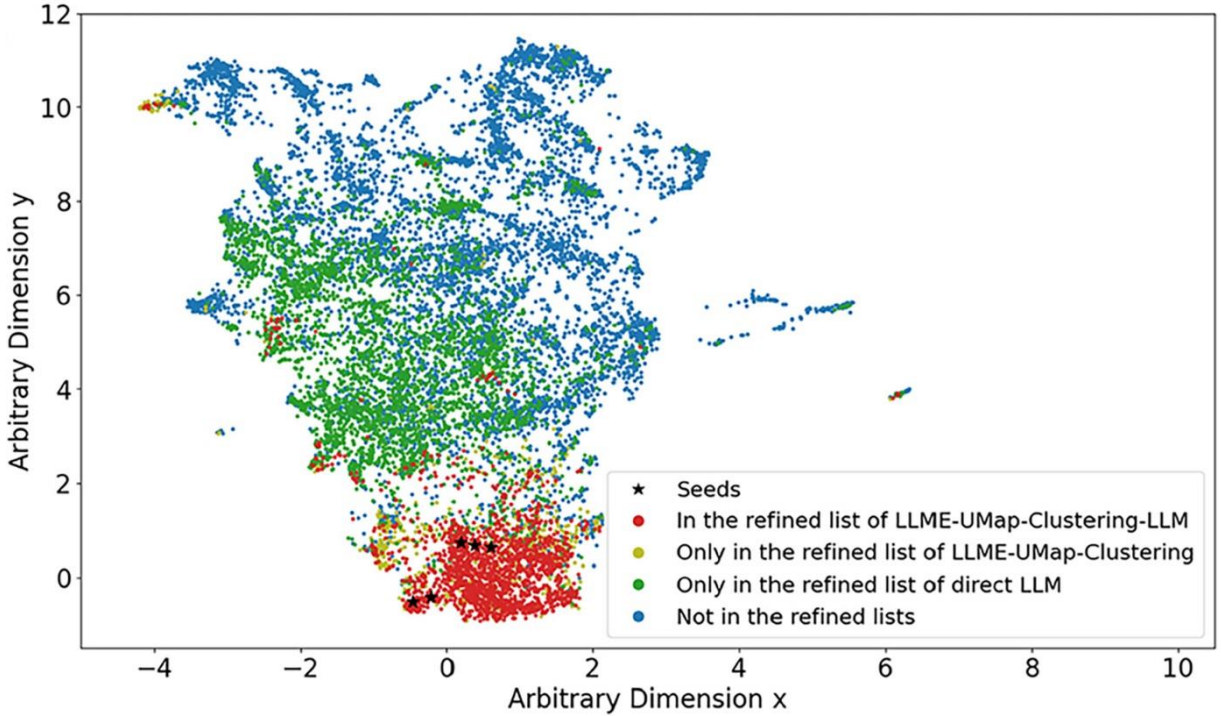
3. Clustering

UMap + K-Means to group similar papers

4. LLM Ranking

Direct GPT-4 queries to score relevance

Visualization



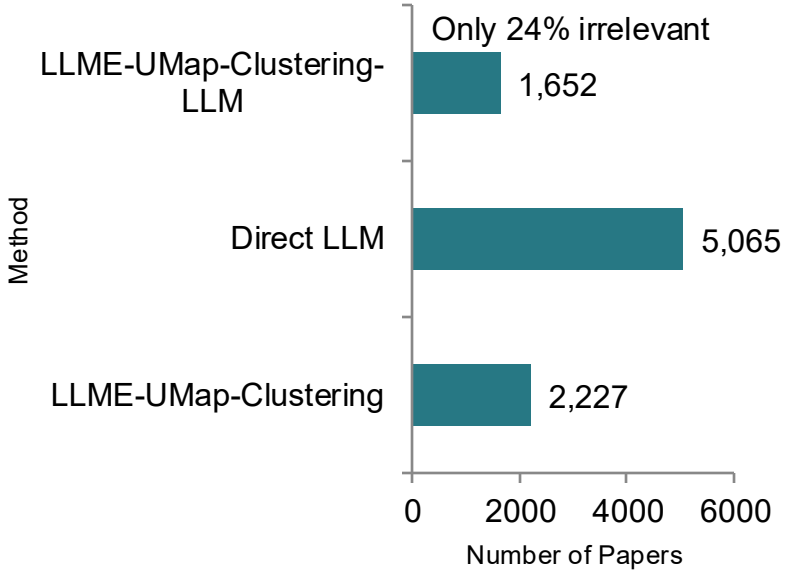
Article Retrieval: Performance Comparison

Best Approach

Combined **LLME-UMap-Clustering-LLM method**

- 88.2% recall of benchmark papers
- Only 13.3% of total search results
- Cost: Only $1.96 for processing 12,431 papers

Importance: Helps with lack of bulk download options from API



Step 2: Text Mining with GPT-4 Turbo

Extracting Material Properties

GPT-4 Turbo extracts structured data from unstructured **text** and **tables** in PDF documents.

Tools Used

- Azure **Document Intelligence** for PDF parsing (Nougat)
- **GPT-4 Turbo** with 128K token context window
- **Function calling** for structured outputs (*IMP*)

Properties Extracted

- Power conversion efficiency (PCE)
- Open circuit voltage (VOC)
- Short circuit current density (JSC)
- Fill factor (FF)

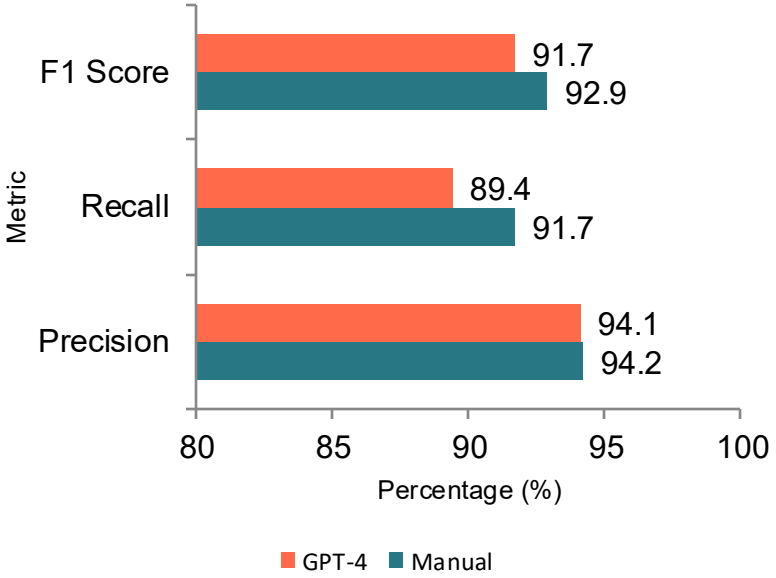
Text Mining: Accuracy Results

Comparable to Human Performance

GPT-4 achieved accuracy nearly equivalent to manually curated datasets.

- Precision: 94.1% (vs. 94.2% manual)
- Recall: 89.4% (vs. 91.7% manual)
- F1 Score: 0.917 (vs. 0.930 manual)

Note: **Less effective** at extracting data from tables compared to plain text



Step 3: Image Mining for SMILES Extraction

Converting Molecular Images to SMILES

SMILES notation enables computational analysis of molecular structures.

Step 1: Object Detection

Azure **Custom Vision** identifies and crops molecular structures from publication figures

Precision: 83.8% | Recall: 86.9%

Step 2: SMILES Conversion

Evaluated 4 ML tools: **MolScribe**, **DECIMER**, Img2Mol, SwinOCSR

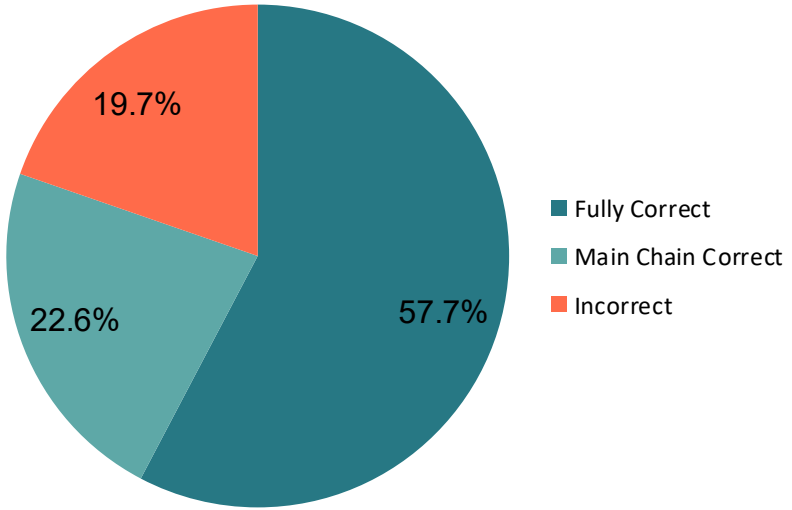
Best: MolScribe (80.3% accuracy)

Image Mining: Results Comparison

MolScribe Performance

- 57.7% fully correct structures
- 22.6% correct main chain (side chain errors)
- 19.7% incorrect recognition

Common errors: low resolution images, non-typical elements, complex structures with 5+ rings



Case Study: OPV Materials Database

Validation Dataset

Manually curated benchmark of organic photovoltaic donor materials

- 503 research papers
- 1,203 OPV donor materials
- 12 material properties
- 1 year to build manually

AI Workflow Results

Applied semi-automated workflow to the same research domain

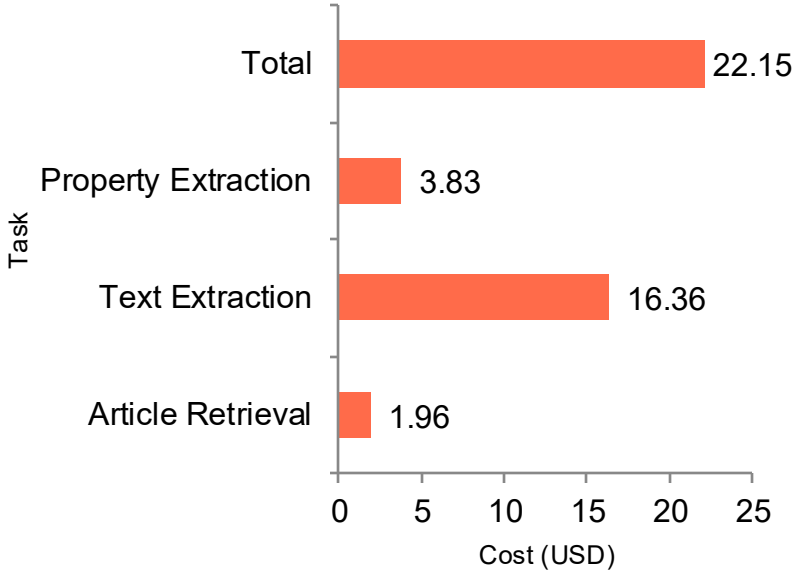
- 88.2% of papers retrieved
- 94% accuracy in data extraction
- Completed in **days**, not months
- **Reasonable cost**

Cost-Effectiveness Analysis

Dramatically Lower Costs

Total Processing Cost

- Article retrieval: $1.96 (~ 12,000 Articles)
- Text extraction: $16.36
- Property extraction: $3.83
- Total: ~$22 for 200+ papers



Key Advantages of the Workflow

Efficiency

Reduces dataset construction from months to days

Accuracy

Performance comparable to human experts, even catching manual errors

Scalability

Adaptable across research domains with minimal customization

Accessibility

Uses affordable cloud APIs and open-source tools (YOLO – object detection)

Criticism

Incomplete

Yet to link chemical structures to material names/IDs

Commercial Ecosystem

Dependency on Microsoft Ecosystem

Dubious costs

Cost is only for 200 papers not for all the papers in the benchmark set

Comparison

Manual extraction involved 12 parameters, whereas this automated workflow only extracted 4.

Thank You